

Population Distributions of Footbridge Modal Properties: A PRISMA-Guided Review and Hierarchical Bayesian Analysis

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Abstract

Keywords:

1. Introduction

The increasing adoption of light weight materials and slender structural forms has made modern footbridges more susceptible to human-induced vibrations. As a result, vibration serviceability under pedestrian loading has become an important design consideration to ensure acceptable structural performance and user comfort. Such assessment depends on several aspects of the bridge, including its structural form, material properties, and dynamic characteristics [1, 2, 3]. Therefore, prior knowledge or reliable estimation of key modal parameters is essential at the design stage, as these parameters provide an initial basis for assessing the feasibility of a proposed design [4, 5, 6, 7].

The current vibration serviceability codes and guidelines represent pedestrian loading through simplified load models to enable practical application at the design stage [8, 9, 10, 11, 12, 13, 14]. Since these load models are used to assess the vibration response of a wide range of footbridges, their adequacy should be evaluated in terms of the resulting structural response, rather than solely through how accurately they reproduce pedestrian loading.

The key limitation of many existing simplified load models is that pedestrians are represented as moving forces, with the corresponding load models calibrated without explicitly accounting for the dynamic interaction between the pedestrians and the supporting structure [15, 16, 17]. Human-structure interaction (HSI) alters the dynamic properties of the coupled pedestrian-structure system, and can lead to structural responses that differ substantially from those predicted by simplified load models. These effects can be

31 particularly important for lightweight structures and for cases with relatively
32 large pedestrian-to-structure mass ratios [18, 17]. Consequently, simplified
33 code models calibrated without accounting for HSI may exhibit systematic
34 response bias that varies with the dynamic characteristics of the bridge, mo-
35 tivating their reassessment and response-based recalibration.

36 Response-based calibration provides such a framework by comparing the
37 response predicted by the simplified design model with that obtained from ex-
38 perimental observations or a higher-fidelity reference model over a prescribed
39 range of structural properties. Previous studies have adopted this approach
40 by evaluating simplified pedestrian loading or vibration serviceability mod-
41 els over ranges of natural frequency, damping and other structural param-
42 eters [19, 3, 17, 18, 20]. However, these structural domains considered in
43 such studies have generally been defined using prescribed parameter ranges,
44 idealised structural configurations, or regularly sampled parameter combi-
45 nations rather than using the observed statistical distribution of dynamic
46 properties within the footbridge population. As a result, some structural
47 conditions may receive disproportionate importance in the calibration sim-
48 ply because of the adopted sampling scheme, rather than because they are
49 common in practice.

50 For code-model calibration, it is important to know not only the range
51 of structural properties that may occur, but also how common these proper-
52 ties are within the footbridge population. Several recent studies have made
53 valuable efforts to compile databases of footbridge characteristics and modal
54 properties. Oliveira et al. (2014) [21] reported fundamental frequencies
55 for 79 footbridges in Portugal, while Van Nimmen (2015) reviewed data
56 for more than 160 footbridges [22, 23]. Wei et al. (2019) [24] assembled
57 dynamic-property data for 138 footbridges, and Gong et al. (2024) [25]
58 more recently studied a collection of footbridges in China. These studies
59 provide a substantial starting point, but their datasets were assembled for
60 different purposes, report different properties, and include some of the same
61 bridges. Further observations are dispersed across many individual case stud-
62 ies [26, 6, 27, 20]. Consolidating these records, identifying duplicate bridges,
63 and recording which modal properties are available would provide a clearer
64 basis for estimating their observed distributions.

65 This study addresses the above gap in two stages. First, a systematic
66 literature search and screening procedure following the PRISMA framework
67 is used to identify relevant publications and compile bridge-level data on key
68 modal properties. Second, hierarchical Bayesian regression is used to proba-
69 bilistically characterise incompletely observed modal properties while retain-
70 ing the uncertainty associated with their estimation. The resulting database
71 provides a population-level description of key footbridge modal properties

and their relationships, allowing the prevalence of different combinations of natural frequency, damping ratio and modal mass to be quantified. Such information provides a statistical basis for distinguishing commonly occurring structural conditions from those that are comparatively rare. This, in turn, can support the selection and weighting of representative structural parameter combinations used in vibration-serviceability assessment and code-model calibration.

Accordingly, this study aims to characterise the modal properties of the footbridge population using a systematically assembled bridge-level database of approximately 250 structures. The principal contributions of the study are: (i) the systematic identification and consolidation of published footbridge dynamic-property data; (ii) the construction of a large, non-duplicated database of bridge-level modal properties; (iii) the probabilistic treatment of incompletely observed damping and modal-mass information using hierarchical Bayesian regression; and (iv) the population-level characterisation of modal properties relevant to vibration-serviceability assessment and code-model calibration.

2. Systematic database development

The footbridge database was developed through a systematic literature identification and screening process following the PRISMA framework [28]. The procedure includes publication identification, title and abstract screening, full text eligibility assessment, bridge level data extraction and consolidation of information into a single database. The following subsections describe the search and screening procedure, the data extraction process, and the resulting database.

2.1. Review scope and search strategy

The systematic review aimed to identify published investigations reporting structural and dynamic properties of full scale pedestrian bridges. The principal variables of interest were bridge span, deck width, fundamental vertical natural frequency, vertical damping ratio, Modal mass, structural form, cross section, and construction materials.

Scopus was selected as the literature database because of its broad coverage of peer-reviewed engineering research [29]. The candidate publications were identified using Elsevier Scopus search API through title, abstract, and keyword fields. The search used a broader combination of pedestrian-bridge terminology with terms relating to natural frequency, damping, and structural mass, experimental dynamic testing, ambient vibration testing, operational modal analysis, model updating and vibration serviceability.

110 The search returned 727 bibliographic records before deduplication. Dub-
111 plicate records were identified using DOI, Scopus EID. Following deduplica-
112 tion, 697 unique records were retained for screening. A DOI was available for
113 529 records, while Scopus EID were retained for publications without a DOI.
114 Scopus records were used to identify and screen publications, while numerical
115 bridge properties were extracted from the full texts of the eligible studies.

116 *2.2. Eligibility criteria*

117 Publications were considered eligible when they reported an investigation
118 of an existing, full-scale pedestrian bridge and contained experimentally mea-
119 sured or identified structural or modal properties that could be associated
120 with a specific structure. Studies based only on laboratory specimens, scaled
121 models, hypothetical structures, or purely numerical investigations were ex-
122 cluded. Publications were also excluded where the reported frequency could
123 not be identified as a structural vertical natural frequency relevant to the
124 present study. For inclusion in the bridge-level database, sufficient informa-
125 tion was required to identify the bridge and extract its fundamental vertical
126 natural frequency together with the geometric information required for the
127 population analysis. Damping ratio, modal mass, structural form, cross-
128 section, and construction material were extracted where available but were
129 not mandatory for inclusion.

130 *2.3. Study screening and bridge identification*

131 Following bibliographic deduplication, the 697 unique publications were
132 retained for full-text screening. The full-text documents were processed
133 using a semi-automated Python workflow. PDF text was extracted using
134 PyMuPDF and subsequently searched for evidence of full-scale pedestrian
135 bridge investigations and relevant structural and modal properties.

136 The extracted text was searched for bridge names and terminology as-
137 sociated with natural frequency, damping, modal mass, mode shapes, and
138 experimental vibration testing. Reference sections were excluded from the
139 search to reduce the likelihood of identifying bridges mentioned only in cited
140 literature.

141 During the screening stage, each publication was counted as a single
142 record, regardless of how many bridges it investigated. During data ex-
143 traction, however, each individual bridge was treated as a separate database
144 record. Consequently, a single publication could contribute data for several
145 bridges.

146 Each detected bridge was assigned a separate candidate record containing
147 its source publication, bridge identity, and associated structural and modal-
148 property evidence. Candidate records were subsequently reviewed against

the bridge-level inclusion requirements defined in Section 2.2 before consolidation.

2.4. Bridge-level data extraction and consolidation

For each candidate bridge, the extracted information included bridge name and location, span, deck width, fundamental vertical natural frequency, vertical damping ratio, modal mass, structural form, cross-section, and construction material. Numerical quantities were converted to consistent units before consolidation.

Because the same physical bridge could be reported in several publications, repeated bridge records were identified using bridge name, location, geometry, structural characteristics, reported modal properties, and source references. Potential matches across different publications were reviewed manually rather than being automatically removed.

Where several publications referred to the same physical bridge, the available information was consolidated into a single bridge-level record. The original experimental investigation was retained as the principal source where available, while additional information from later publications was incorporated where appropriate.

2.5. Resulting database and data completeness

Following bridge-level screening and duplicate reconciliation, 250 unique footbridges were retained in the final database. Fundamental vertical natural frequency and span were available for all bridges, while deck width was available for 240 bridges. Vertical damping ratios were reported for 100 bridges, whereas mass related information was considerably less complete, with modal mass information available for only 30 bridges and linear mass available for 25 bridges. Structural form and material information were available for all bridges, while cross-section information was available for 242 bridges. The resulting differences in data completeness, particularly for damping and mass-related properties, motivated the probabilistic treatment introduced in the following section.

3. Bayesian treatment of incomplete bridge data

The database contains complete records of fundamental vertical natural frequency and span, but damping and mass-related properties are reported for only a subset of bridges. Analysing only bridges with complete records would therefore exclude a substantial proportion of the database. To retain these bridges in the population analysis, Bayesian regression models were

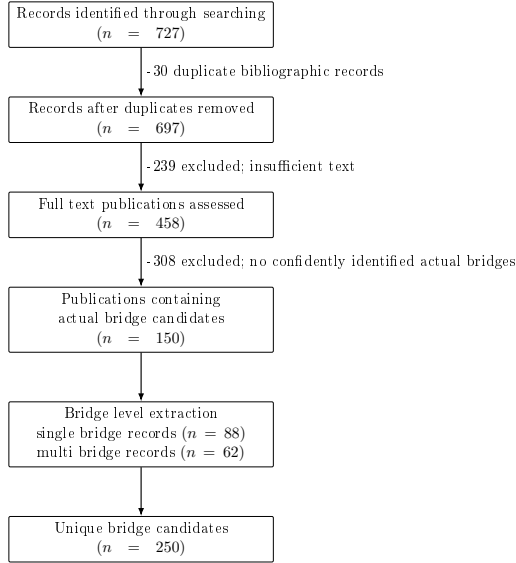


Figure 1: PRISMA flow diagram for the systematic identification, screening, and consolidation of footbridge data.

185 developed to infer the unreported properties from the available bridge char-
 186 acteristics. The missing values were represented by posterior predictive dis-
 187 tributions, allowing uncertainty in both regression parameters and remaining
 188 variability between bridges to be propagated into resulting population dis-
 189 tributions.

190 3.1. Hierarchical model for vertical damping ratio

191 Let ξ_i denote the damping ratio of the first vertical mode for the i -th
 192 bridge. The damping ratios ξ_i differ between bridges, reflecting differences
 193 in their construction materials, structural forms, and other factors. How-
 194 ever, the database does not provide sufficient details to model the physical
 195 mechanisms governing damping in each bridge. Statistical modelling offers a
 196 means of estimating unreported values from observed data without explicitly
 197 identifying the underlying mechanisms [30, 31, 32]. Accordingly, a Bayesian
 198 hierarchical regression model was fitted to the reported damping ratios, us-
 199 ing natural frequency, span, and material as predictors that are available in
 200 the database. The fitted model was then used to obtain posterior predictive
 201 distributions for unreported damping ratios, accounting for both uncertainty
 202 in the regression parameters and remaining variability between bridges. The
 203 probabilistic structure of the hierarchical damping model is shown in Fig. 2.

204

205 For each bridge, the available predictor information is $x_i = \{f_i, L_i, m_i\}$,

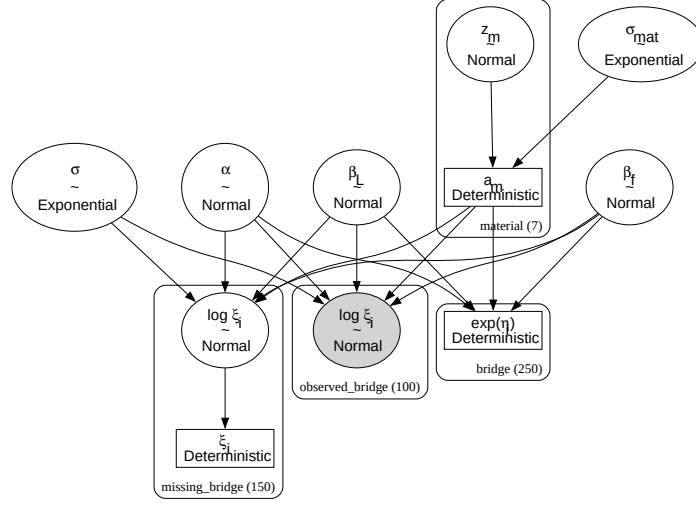


Figure 2: Probabilistic structure of the hierarchical Bayesian model for vertical damping ratio.

where f_i is the fundamental vertical natural frequency, L_i is the span, and m_i denotes the material category. Let $\mathcal{O}$ and $\mathcal{M}$ denote the sets of bridges with observed and missing damping ratios, respectively. The data used to estimate the model parameters are then

$$\mathcal{D}_{\text{obs}} = \{(\xi_i, \mathbf{x}_i) : i \in \mathcal{O}\}. \quad (1)$$

Damping ratios were modelled using a lognormal distribution to ensure positive predictions, with the corresponding log damping ratios assumed to follow a normal distribution:

$$\log \xi_i \mid \eta_i, \sigma \sim \mathcal{N}(\eta_i, \sigma^2), \quad (2)$$

where η_i is the conditional mean of log damping ratio and σ describes the remaining variability between bridges. The conditional mean was expressed as

$$\eta_i = \alpha + \beta_f x_{f,i} + \beta_L x_{L,i} + a_{m_i}, \quad (3)$$

where α is the common intercept, β_f and β_L describe the associations with natural frequency and span, and a_{m_i} is the effect of the material category of bridge i . The predictors f_i and L_i were logarithmically transformed and standardised as,

$$x_{f,i} = \frac{\log f_i - \overline{\log f}}{s_{\log f}}, \quad x_{L,i} = \frac{\log L_i - \overline{\log L}}{s_{\log L}}. \quad (4)$$

Here, the overbars and s denote the means and standard deviations of the corresponding log-transformed predictors.

The material categories contain different numbers of observed damping ratios. Estimating an independent model for each category would therefore provide limited information for sparsely represented materials. Instead, material effects were assigned a shared distribution,

$$a_m \mid \sigma_{\text{mat}} \sim \mathcal{N}(0, \sigma_{\text{mat}}^2). \quad (5)$$

This hierarchical structure allows information to be partially pooled across materials while retaining differences between categories. The parameter σ_{mat} controls the variation in material effects around the common baseline.

Let $\boldsymbol{\theta}$ contain the regression coefficients, material effects, and dispersion parameters:

$$\boldsymbol{\theta} = \{\alpha, \beta_f, \beta_L, \mathbf{a}, \sigma, \sigma_{\text{mat}}\}. \quad (6)$$

Assuming conditional independence between bridge records, the likelihood of the observed log damping ratios is

$$\mathcal{L}(\mathcal{D}_{\text{obs}} \mid \boldsymbol{\theta}) = \prod_{i \in \mathcal{O}} \mathcal{N}(\log \xi_i \mid \eta_i, \sigma^2). \quad (7)$$

The posterior distribution follows from Bayes' rule:

$$p(\boldsymbol{\theta} \mid \mathcal{D}_{\text{obs}}) \propto \mathcal{L}(\mathcal{D}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}). \quad (8)$$

For a bridge with unreported damping, its known predictors were used to obtain the posterior predictive distribution:

$$p(\xi_i \mid \mathbf{x}_i, \mathcal{D}_{\text{obs}}) = \int p(\xi_i \mid \mathbf{x}_i, \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathcal{D}_{\text{obs}}) d\boldsymbol{\theta}, \quad i \in \mathcal{M}. \quad (9)$$

This distribution accounts for both uncertainty in the model parameters and the remaining variability between bridges. Samples from these distributions were used to represent the missing damping ratios in the subsequent population analysis.

3.2. Bayesian estimation of pedestrian-to-structure mass-ratio probabilities

The pedestrian-to-structure mass ratio depends on both the bridge modal properties and the distribution of pedestrians. The present analysis therefore considers the prevalence of mass ratios at a specified pedestrian density ρ .

For bridge i , assuming pedestrians are uniformly distributed over the occupied deck area A_i , the pedestrian modal mass associated with the first vertical mode is

$$M_{p,i}^*(\rho) = \rho \bar{m}_p \int_{A_i} \phi_i^2(\mathbf{x}) dA, \quad (10)$$

247 where $\bar{m}_p$ is the assumed mean pedestrian mass and ϕ_i is the corresponding
 248 mode shape. The pedestrian-to-structure modal mass ratio is then,

$$\mu_i(\rho) = \frac{M_{p,i}^*(\rho)}{M_{b,i}^*}, \quad (11)$$

249 where $M_{b,i}^*$ is the bridge modal mass evaluated using the same mode-shape
 250 ϕ_i .

251 Sufficient information to calculate modal mass ratios is available for only
 252 a subset of bridges in the database. A Bayesian model was therefore used
 253 to estimate the proportion of bridges whose mass ratios fall below a spec-
 254 ified threshold T , without estimating a missing mass ratio for each bridge.
 255 Threshold probabilities were estimated separately for each material category
 256 and then combined according to the estimated prevalence of these categories.
 257 The probabilistic structure of the Bayesian mass-ratio model is shown in Fig-
 ure 3.

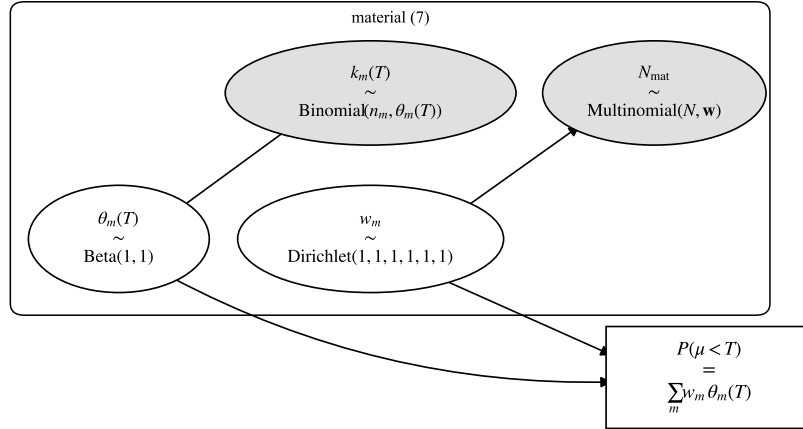


Figure 3: Probabilistic structure of the Bayesian model for pedestrian-to-structure mass-ratio probabilities.

258 For a material category m , let $\theta_m(T; \rho)$ be,
 259

$$\theta_m(T; \rho) = P(\mu(\rho) < T \mid m). \quad (12)$$

260 Let n_m be the number of bridges with calculable mass ratios in category m ,
 261 and let $k_m(T; \rho)$ be the number of these bridges whose mass ratios are below
 262 the threshold T . The first likelihood related these counts to the material
 263 specific threshold probability:

$$k_m(T; \rho) \sim \text{Binomial}(n_m, \theta_m(T; \rho)), \quad (13)$$

264 This formulation assumes that bridges with calculable mass ratios are repre-
 265 sentative of their material category with respect to mass ratio.

266 The second likelihood describes the material composition of the full
 267 database. Let N_m be the total number of bridges in material category m ,
 268 including those without calculable mass ratios, and let $w_m = P(m)$ denote
 269 the corresponding population proportion. For M mutually exclusive and
 270 exhaustive material categories,

$$N_{mat} \mid w \sim \text{Multinomial}(N, w), \quad (14)$$

271 where $N_{mat} = (N_1, \dots, N_M)$, $w = (w_1, \dots, w_M)$, $N = \sum_{m=1}^M N_m$, and $\sum_{m=1}^M w_m =$
 272 1. Consequently, all bridges contribute to estimating the material propor-
 273 tions, while only bridges with calculable mass ratios contribute to estimating
 274 the material-specific threshold probabilities. The prior distributions for these
 275 parameters are specified in the following subsection.

276 The material proportions and threshold probabilities were combined using
 277 the law of total probability:

$$\begin{aligned} P(\mu(\rho) < T) &= \sum_{m=1}^M P(m) P(\mu(\rho) < T \mid m) \\ &= \sum_{m=1}^M w_m \theta_m(T; \rho). \end{aligned} \quad (15)$$

278 Each product gives the probability that a bridge belongs to material category
 279 m and has a mass ratio below T . Summing across the material categories
 280 give the overall proportion of bridges below the threshold.

281 To propagate uncertainty in both components, the overall probability was
 282 evaluated for each posterior sample s :

$$F_{\mu}^{(s)}(T; \rho) = \sum_{m=1}^M w_m^{(s)} \theta_m^{(s)}(T; \rho). \quad (16)$$

283 The resulting samples account for uncertainty in both the material propor-
 284 tions and the material-specific threshold probabilities.

285 The calculation was repeated over a range of thresholds T at a specified
 286 pedestrian density ρ to characterise the cumulative distribution of pedestrian-
 287 to-structure mass ratio. At each threshold, the posterior mean and 95%
 288 credible interval were calculated for $P(\mu(\rho) < T)$. Lower and upper bound
 289 of these pointwise credible intervals define the bounding curves of a proba-
 290 bility box (p-box), allowing the estimated proportion of bridges below any
 291 selected mass ratio threshold to be assessed together with its uncertainty.

292 3.3. Prior specification and posterior computation

293 As described in the Section 3.1, the damping model contains 12 unknown
 294 model parameters, including the intercept α , two regression coefficients β_f
 295 and β_L , the residual standard deviation σ , and the between-material standard
 296 deviation σ_{mat} . The remaining seven parameters are the material-specific ef-
 297 fects a_m , corresponding to the seven material categories represented in the
 298 database: steel, concrete, composite, fibre-reinforced polymer (FRP), alu-
 299 minium, timber, and other materials. Unreported damping ratios are addi-
 300 tional latent quantities whose distributions are inferred from the model.

301 The intercept and regression coefficients were assigned normal priors:

$$\alpha \sim \mathcal{N}(\log(0.01), 1^2), \quad \beta_f, \beta_L \sim \mathcal{N}(0, 0.5^2), \quad (17)$$

302 where the second argument denotes the variance. The prior for α was cen-
 303 tred at $\log(0.01)$, corresponding to a baseline damping ratio of 1%. This
 304 value was selected to place the prior within the range commonly associated
 305 with footbridge vibration serviceability, where damping ratios are generally
 306 between 0.4% to 2% [9, 13, 14] for vibrations under pedestrian excitaions
 307 and vary with construction material. The relatively broad prior on the loga-
 308 rithmic scale allows the observations and material specific effects to shift the
 309 baseline towards both lower and higher values.

310 The regression coefficients were assigned zero-mean priors, reflecting the
 311 absence of prior knowledge about the direction of their effects and to avoid
 312 favouring any particular relationship between damping and the predictors.
 313 Because the log-transformed predictors were standardised, the same prior
 314 scale was used for both coefficients, regularising large effects while allowing
 315 the observations to inform their direction and magnitude.

316 The material-specific effects were assigned a shared normal prior,

$$a_m \mid \sigma_{\text{mat}} \sim \mathcal{N}(0, \sigma_{\text{mat}}^2). \quad (18)$$

317 This prior treats the material effects as deviations from the common re-
 318 gression relationship, centered at zero. Sharing the same distribution across
 319 material categories allows partial pooling, so that material categories with
 320 few observations can borrow information from those with more data, while
 321 still allowing for differences between categories. The between-material stan-
 322 dard deviation controls the extent of variation among the material effects
 323 and was assigned the prior

$$\sigma_{\text{mat}} \sim \text{Exponential}(4). \quad (19)$$

324 An exponential prior was used because σ_{mat} is a positive scale parameter.
 325 This prior restricts σ_{mat} to positive values and places greater prior probability

on relatively small differences between material categories while retaining support for larger differences if required by the data.

The remaining bridge-to-bridge variability not explained by the predictors or material effects is represented by the residual standard deviation σ , for which the prior

$$\sigma \sim \text{Exponential}(2) \quad (20)$$

was adopted. The exponential prior similarly restricts the residual standard deviation to positive values while allowing substantial unexplained variability between bridges.

For computation, the material effects were expressed using the non-centred parameterisation

$$z_m \sim \mathcal{N}(0, 1), \quad a_m = \sigma_{\text{mat}} z_m. \quad (21)$$

For the mass-ratio model described in Section 3.2, prior distributions were required for the material proportions w_m and the material-specific threshold probabilities $\theta_m(T; \rho)$. The material proportions were assigned a symmetric Dirichlet prior,

$$w \sim \text{Dirichlet}(1, \dots, 1), \quad (22)$$

where the number of concentration parameters corresponds to the number of material categories considered in the mass-ratio analysis. This prior treats the material categories symmetrically and does not favour any particular composition of the bridge population before observing the material counts in the database.

The material-specific threshold probabilities were assigned independent uniform beta priors,

$$\theta_m(T; \rho) \sim \text{Beta}(1, 1). \quad (23)$$

This prior assigns equal density to all probabilities between zero and one and therefore does not favour a particular probability that the mass ratio falls below the selected threshold within any material category. The same prior specification for $\theta_m(T; \rho)$ was applied independently at each threshold T , whereas the material proportions w describe the common material composition of the database and are therefore unchanged across thresholds.

Both Bayesian models were implemented in PyMC and posterior sampling was performed using the No-U-Turn Sampler (NUTS), an adaptive Hamiltonian Monte Carlo algorithm. For each model, four independent chains were run with 2000 tuning iterations followed by 2000 retained posterior samples per chain, using a target acceptance probability of 95%.

358 3.4. *Model assessment procedures*

359 4. Results

360 4.1. *Database characteristics and observed distributions*

361 4.2. *Population distribution of vertical damping ratio*

362 4.3. *Population distribution of pedestrian-to-structure mass ratio*

363 4.4. *Joint prevalence of serviceability-relevant properties*

364 5. Discussion

365 6. Conclusions

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